Introduction and Methodology

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ECON 8260
Spring 2026

Readings

- None required but useful background:
  - Lewbel (2019), esp. Sections 1, 3.1-3.5, 5
  - *Mostly Harmless Econometrics* by Angrist and Pischke
  - *Causal Inference: The Mixtape* by Cunningham

Identification

Lewbel (2019)

- Model parameters are *uniquely determined* from observable population that generated the data
- Let θ denote unknown parameters to be estimated (e.g., utility function parameters, regression coefficients)
- For θ to be identified, alternative values of θ must imply different distributions of the observable data
- If θ is not identified, we cannot find a consistent estimator for θ
- Recognizing lack of identification and making (and justifying) restrictions and assumptions that suffice to attain it are key in econometric modeling
  - E.g., exclusion restrictions, functional form assumptions, distributional assumptions, assumed sources of exogenous variation

Rubin's Model of Causation

- We are interested in:
$$Y_i(1) - Y_i(0)$$
where $Y_i(1) = Y_i(T_i = 1)$ for some treatment T and unit i
- $Y_i(1)$ and $Y_i(0)$ are potential outcomes; that is, outcomes for i had they received the treatment or control, respectively
  - Treatment examples: Taking sickness leave, using a medication, being exposed to pollution in utero
- Fundamental problem of causal inference: only $Y_i(1)$ or $Y_i(0)$ is observed for $i \rightarrow$ can never find true causal effect of treatment T for unit i

Rubin's Model of Causation

- Instead of the individual treatment effect, we might be interested in the average treatment effect (ATE):

$$E[Y(1) - Y(0)] = E[Y(1)] - E[Y(0)]$$

- Cannot find ATE b/c $Y(1)$ and $Y(0)$ are not observed for all units in the population
- Might also be interested in average treatment effect on the treated (ATT):

$$E[Y(1|T = 1) - Y(0|T = 1)]$$

- Cannot find ATT b/c $Y(0|T = 1)$ is never observed

Rubin's Model of Causation

- Although we cannot observe true ATE, we might estimate it as follows:

$$\begin{aligned}\widehat{ATE} &= E[Y(1) - Y(0)] \\ &= E[Y(1)] - E[Y(0)] \\ &= E[Y_t(1)] - E[Y_c(0)]\end{aligned}$$

  where $Y_t(\,)$ and $Y_c(\,)$ are potential outcomes for treated and control units, respectively

- Calculate average Y for observations that received treatment and average Y for observations that did not (control group)
- Yields an unbiased estimate of the ATE under two assumptions:
  - Stable unit treatment value assumption (SUTVA)
  - Unconfoundedness/ignorability

Rubin's Model of Causation

SUTVA

- Treatment status of any unit does not affect potential outcomes of *other* units
  - Violation example: Employer-sponsored vaccination mandate for nursing home (NH) staff may impact non-mandate NHs via declines in local transmission or cross-NH employment
- Treatments for all units are comparable (i.e., no variation in treatment)
  - Violation example: Treatment is hospital admission, but hospitals vary considerably in quality

Rubin's Model of Causation

Unconfoundedness (strong ignorability)

$$(Y(1), Y(0)) \perp T$$

- Treatment assignment is independent of the outcomes Y(1) and Y(0)
- Violations:
  - Omitted variable bias (OVB): Treated units differ on some unobserved variable that affects outcomes (e.g., conscientiousness, future orientation)
  - Selection: Treated units have higher expected treatment effects (e.g., return to college is highest for those that go, effect of knee replacement surgery on physical mobility is highest for those who do it)

Estimating ATE with Randomized Experiments

- Unconfoundedness assumption unlikely to hold when units choose T. What do we do?
- Randomized experiment! Often considered "gold standard"
  1. Randomly sample units from population
  2. Randomly assign treatment and control to the units
  3. Estimate ATE $= \widehat{ATE} = E[Y_t(1)] - E[Y_c(0)] =$ ATT
  4. In regression terms, simply estimate:

     $$Y_i = \beta_0 + \beta_1 T_i + u_i$$

     where T_i is the randomly assigned treatment
- Simply compare differences in outcomes among treated and control groups

Estimating ATE with Randomized Experiments

- Yields an unbiased estimate of the causal effect of treatment (ATE). Why?
- Random assignment guarantees:
  - Treated and control groups are on average the same (no OVB)
  - Treated units cannot have higher expected treatment effects (no selection)
  - In regression terms, $E[u|T] = 0 \rightarrow$ Treatment is uncorrelated w/ all other factors that determine Y
- But, randomized experiments are not a panacea:
  - In the absence of random sampling, need random application (not the same as random assignment)
  - Need random dropouts; more generally, might worry about compliance w/ treatment
  - May be concerns about general equilibrium effects

Estimating ATE with Randomized Experiments

- If randomized experiments can identify causal effects of treatment, why not always take this approach?
  - Often not feasible. E.g., we cannot randomize merger of two large hospitals, a person's birth weight
  - Expensive and difficult to do (need significant administrator buy in)
  - Ethical concerns regarding exposing a group to a harmful intervention or excluding a group from a beneficial one
- Ideally, would like experimental data to understand economic relationships
- But, in most cases we have observational data
- To find answers, we need econometric skills and creativity

Furious Five

- Five essential econometric tools to uncover causal effects
  1. Randomized Assignment
  2. Regression / Selection on Observables
  3. Difference-in-Differences (DD)
  4. Instrumental Variables (IV)
  5. Regression Discontinuity (RD)
- DD, IV, RD typically identify causal effects from "natural experiments"
- Natural experiment: Random variation in treatment that stems from variation in policies, events, circumstances, or accidents of nature
  - Generates exogenous variation in variables that are otherwise endogenous

Difference-in-Differences (DD)

Preliminaries

- Compare outcomes of treated group (those affected by policy/event) before and after the policy/event relative to outcomes of a control group (who did not experience policy/event) during the same period
- B/c units are observed before and after treatment, can eliminate *time-invariant unobserved differences* b/w treatment and control groups

Difference-in-Differences (DD)

Garthwaite et al. (2014)

- Exploit change in Tennessee's Medicaid system to estimate effect of public health insurance eligibility on labor supply of childless adults
  - In 2005, TN discontinued expansion of Medicaid $\rightarrow$ $\sim$200K adults abruptly lost coverage
- Compare employment of childless adults in TN to employment of childless adults in KY after (2006) vs. before disenrollment (2004)

Difference-in-Differences (DD)

Garthwaite et al. (2014)

- Estimate:

$$Y_{st} = \alpha + \gamma \mathbb{1}(s = TN) + \lambda \mathbb{1}(t = 2006) + \delta \mathbb{1}(s = TN)\mathbb{1}(t = 2006) + \varepsilon_{st}$$

  where Y_{st} is share of childless adults working in state s in year t

- Note:

$$E[Y_{st}|s = KY, t = 2006] - E[Y_{st}|s = KY, t = 2004] = \lambda$$

  and

$$E[Y_{st}|s = TN, t = 2006] - E[Y_{st}|s = TN, t = 2004] = \lambda + \delta$$

- Thus, the difference-in-differences:

$$\{E[Y_{st}|\, s = TN, t = 2006] - E[Y_{st}|s = TN, t = 2004]\} \\ - \{E[Y_{st}|s = KY, t = 2006] - E[Y_{st}|s = KY, t = 2004]\} = \delta$$

Difference-in-Differences (DD)

Garthwaite et al. (2014)

- Could add more control states and more pre- and post-treatment periods
  - Garthwaite et al. use other Southern states as controls and data from 2000–2007
- Add dummy variables for each state, γ_s, and each year, τ_t. That is, add unit and time fixed effects (i.e., two-way fixed effects)

$$y_{st} = \gamma_s + \tau_t + \delta \mathbb{1}(s = TN)\mathbb{1}(t \geq 2006) + \varepsilon_{st}$$

Difference-in-Differences (DD)

Identifying Assumption: Parallel Trends

- *Trends* in outcomes for treatment and control groups would be the same in absence of the treatment
  - I.e., time effects are *common* across treatment and control groups
- Examine support for this assumption using data w/ multiple pre-treatment periods and estimating an event-study model:

$$y_{st} = \gamma_s + \tau_t + \sum_{\ell=2000, \neq 2005}^{2007} \pi_\ell \mathbb{1}(t = \ell)\mathbb{1}(s = TN) + \varepsilon_{st}$$

  - Want π_ℓ coefficients corresponding to pre-treatment periods to be close to zero, statistically insignificant, and not systematically trending
  - See Rambachan & Roth (2023) for rigorous assessment of parallel trends assumption using event-study estimates Example
- Embeds assumption that there are no unit-specific time-varying unobservables that affect treatment and the outcome

Difference-in-Differences (DD)

More Identifying Assumptions

- Composition of treatment and control groups is stable before and after the policy/event
  - If not, estimated effect reflects changes in composition of treatment and control groups induced by (or occurring at the same time as) treatment
- No anticipation: Units do not change behavior before treatment in response to knowing treatment is coming

Difference-in-Difference-in-Differences (DDD)

Garthwaite et al. (2014)

- Worry DD estimate reflects TN-specific shocks other than Medicaid disenrollment
- There is another candidate control group—other adults, which allows them to estimate a "triple differences" specification:

$$y_{gst} = \gamma_{gs} + \gamma_{gt} + \gamma_{st} + \beta \mathbb{1}\{g = childless\}\mathbb{1}\{s = TN\}\mathbb{1}\{t \geq 2006\} + \varepsilon_{gst}$$

  where y_{gst} is share of demographic group g (childless adults or other adults) that works in state s in year t

- Include all pairwise interactions: demographic group $\times$ state FEs (γ_{gs}); demographic group $\times$ year FEs (γ_{gt}); state $\times$ year FEs (γ_{st})
- Parallel trends assumption: Gap in employment b/w childless adults and other adults would have evolved similarly in TN and other Southern states in the absence of treatment
- See Olden & Møen (2022) and Ortiz-Villavicencio & Sant'Anna (2025) for more on DDD

Difference-in-Differences (DD)

Staggered Timing

- In Garthwaite et al. (2014), only TN is treated, but often multiple units receive treatment at different times ("staggered timing")
- In such cases, historically, we estimated TWFE regressions of the type:

$$y_{it} = \gamma_i + \tau_t + \delta D_{it} + \varepsilon_{it}$$

  where D_{it} is an indicator for whether unit i was treated in period t

- Also estimate event-study models of the type:

$$y_{it} = \gamma_i + \tau_t + \sum_{\ell} \pi_\ell \mathbb{1}(t - E_i = \ell) + \varepsilon_{it}$$

  where E_i is when unit i initially receives treatment. Thus, $\ell = t - E_i$ measures time relative to treatment beginning ("event time" or "relative time")

Difference-in-Differences (DD)

Staggered Timing

- Lots of important and recent innovations for the staggered timing case
- Standard TWFE regressions produce biased estimates if treatment effects vary across treatment cohorts or over time
- TWFE parameter is a weighted combination of all possible 2x2 DD comparisons:
  - Treated vs. never treated
  - Early treated vs. later treated
  - Later treated vs. early treated
- Fundamental problem: Forbidden comparisons of later-treated units vs. earlier (already)-treated units (esp. if treatment effects vary over time)
- See Callaway (2022), Roth et al. (2023), Baker et al. (forthcoming) for surveys and alternative estimators

Instrumental Variables (IV)

Preliminaries

- Interested in effect of T on Y, but concerned T is endogenous
- IV estimation occurs in 2 steps:
    1. Estimate: $T_i = \pi_0 + \pi_1 Z_i + \pi_2 X_i + v_i$
    2. Estimate: $Y_i = \beta_0 + \beta_1 \hat{T}_i + \beta_2 X_i + u_i$
- Z_i is the instrument; assumed to be randomly assigned (i.e., there is exogenous variation in Z)
- $\widehat{T}_i$ is the predicted value of treatment based on Z_i from step 1
- Key assumptions:
    1. Relevance: $\pi_1 \neq 0$; instrument is correlated w/ treatment (also written as $corr(T, Z) \neq 0$)
    2. Exclusion restriction: $corr(Z, u) = 0$; instrument is uncorrelated w/ unobservables that impact Y, and only influences Y through its effect on T

Instrumental Variables (IV)

Angrist (1990)

- Uses Vietnam era draft lottery #s as an IV to estimate effect of military service on earnings
- Lottery #s were randomly assigned but influenced probability of treatment (i.e., military service)
  - Each man was assigned lottery # from 1-365 based on random drawings of birth dates; only those with #s below a certain ceiling were draft eligible
- Sample consists of men born from 1950-1953
- For each birth cohort, regress earnings in subsequent years (1981-1984) on a constant and dummy variable for veteran status, T_i:

$$Y_i = \beta_0 + \beta_1 T_i + \varepsilon_i$$

- Concern that veterans may have lower earnings b/c they tend to have lower values of ε (i.e., $corr(T_i, \varepsilon_i) \neq 0$)

Instrumental Variables (IV)

Angrist (1990)

- Instruments are a constant and a dummy variable for draft eligibility, Z_i, which equals 1 if i's lottery # was below the ceiling

$$T_i = \alpha_0 + \alpha_1 Z_i + \nu_i$$

- Identifying assumptions:
  1. Relevance: Person's # is correlated w/ veteran status, $corr(T_i, Z_i) \neq 0$
  2. Exclusion restriction: Person's # is uncorrelated w/ unobservables and only influences earnings via effect on probability of veteran status, $corr(Z_i, \varepsilon_i) = 0$
- Finds military service reduces annual earnings for white men by 15%

Instrumental Variables (IV)

Keane's Comments on Angrist (1990)

- If there are heterogeneous treatment effects (i.e., β_{1i}), IV identifies effect for compliers—those whose behavior is influenced by the instrument (i.e., those induced to serve by the draft)
  - IV identifies local average treatment effect (LATE), but usually we want ATE, $\beta_1 = E(B_{1i})$

$$LATE = \frac{E(Y_i|Z_i = 1) - E(Y_i|Z_i = 0)}{P(T_i = 1|Z_i = 1) - P(T_i = 1|Z_i = 0)}$$

  - In this case, also need monotonicity (no defiers) assumption—IV (weakly) affects all units in same direction
- Unclear what causes adverse effect of military service on earnings; to inform policy, need to understand mechanisms

Instrumental Variables (IV)

Keane's Comments on Angrist (1990)

- Draft lottery may itself affect behavior and is not a valid instrument
  - Those who drew low #s may experience lower earnings b/c, at least for a time, high level of uncertainty caused them to $\downarrow$ human capital investment
  - Possible a person's lottery # affected human capital investment through effects on probability of military service
  - This effect is separate from effect of service itself and would exist w/in treatment group even if none were ultimately drafted

    $\rightarrow$ overstate the negative effect of military service

Regression Discontinuity (RD)

Preliminaries

- Idea is to exploit knowledge about rules determining treatment
- Treatment determined by whether an observed "assignment" variable (also referred to as "forcing" or "running" variable) exceeds a known cutoff
- Suppose treatment status, D_i, is a deterministic and discontinuous function of X_i:

$$D_i = \begin{cases} 1 & \text{if } X_i \geq c \\ 0 & \text{if } X_i < c \end{cases}$$

  where c is the known threshold or cutoff

- Units w/ X_i just below cutoff are used as counterfactuals for units w/ X_i at or just above it

Regression Discontinuity (RD)

Card et al. (2008)

- Study impact of Medicare eligibility on mortality and hospitalizations
- Individuals become eligible for Medicare precisely on 65th birthday
- RD compares outcomes of those just above and just below age 65
- Jumps in the relationship b/w age (measured in year-quarters) and health outcomes in the neighborhood of cutoff are evidence of a treatment effect
  - Should be no reason, other than Medicare eligibility, for outcomes to be a discontinuous function of age
- In other words, the discontinuous jump in outcomes at c is the causal effect of Medicare eligibility

Regression Discontinuity (RD)

Identifying Assumptions

- Continuity of potential outcomes at the cutoff
  - In absence of treatment, conditional expected potential outcomes are continuous in the running variable at c

$$\lim_{x \downarrow c} \mathbb{E}[Y_i(d) \mid X_i = x] = \lim_{x \uparrow c} \mathbb{E}[Y_i(d) \mid X_i = x], \quad d \in \{0, 1\}$$

- No manipulation of running variable
  - Units cannot manipulate X to ensure treatment status in a way that is correlated w/ potential outcomes

Regression Discontinuity (RD)

Card et al. (2008)

- Assuming relationship b/w age and health outcomes is linear, a simple way of estimating treatment effect is via linear regression:

$$Y_i = \alpha + \delta D_i + X_i \beta + \varepsilon_i$$

  where Y_i is a health outcome and δ is the causal effect of interest

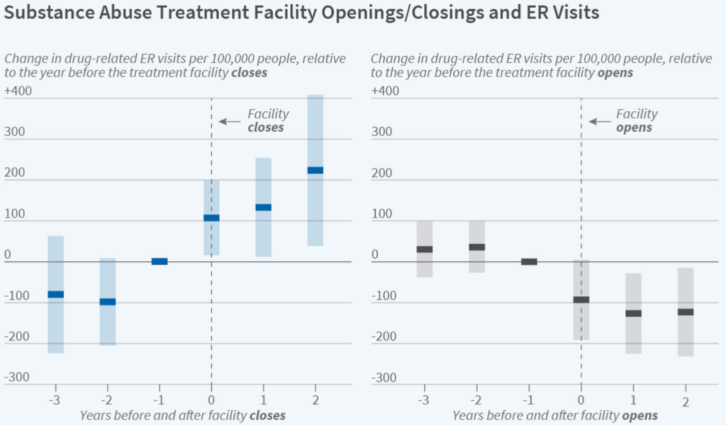
- More flexible specification:

$$Y_i = \alpha + \delta D_i + \beta(X_i - c) + \gamma D_i(X_i - c) + \varepsilon_i$$

- Only a local effect is identified (i.e., for those w/ X_i values around cutoff)
- See Lee & Lemieux (2010) and Cattaneo & Titiunik (2022) for practitioner guides

APPENDIX

Rambachan and Roth (2023)



Openings correspond to the first treatment facility in a geographic area; closures are often in areas with other facilities that remain open. Shaded bars represent 95% confidence intervals. Source: Researchers' calculations using data from the National Directories of Drug and Alcohol Abuse Treatment Centers and the New Jersey Uniform Billing Records

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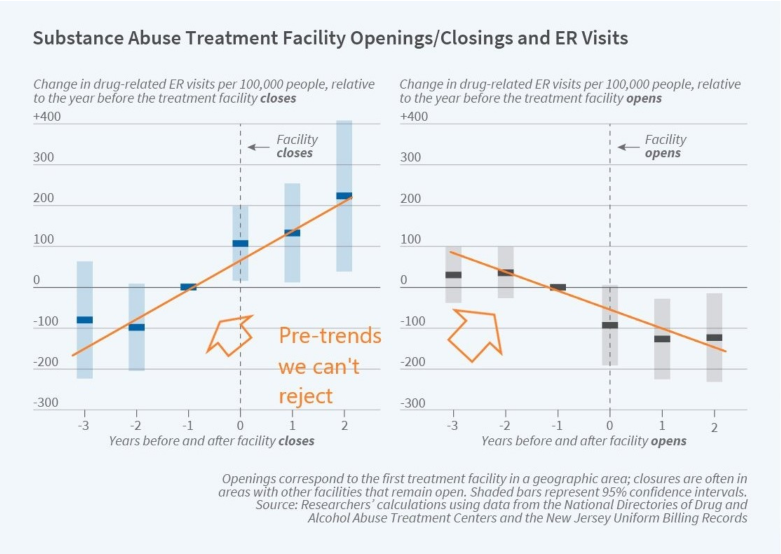
Rambachan and Roth (2023)



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